

Cross-sector links, physical CP and global cap modes

Exact elementary CP, representation-derived alignment and a complete conditional mass spectrum

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The product symmetry removes a quartic deformation. Higher-degree mixed interactions, physical vacuum selection and the global chiral completion remain open.

Concrete advance and remaining boundary

This push constrains the problematic cross-sector interaction by changing the specified family group to $G_u \times G_d$, with each $G_f=3.A6$, and joining the sectors with a bifundamental link. The bare absolute overlap quartic is forbidden. The first dressed overlap is degree six. The link has a unique new holomorphic sextic tensor, whose component ratios come from the previously derived Valentiner invariant. A renormalizable supersymmetric messenger chain generates this contraction at leading tree order, with its complete allowed holomorphic operator audit retained. A positive alignment interaction has exact zero-energy states at every group matrix, without inserting a CKM angle.

We also construct an exact anti-linear involution on the specified triplet and singlet fields. It leaves the sextic conjugate as required, and a canonical quark calculation produces a pair with opposite nonzero Jarlskog invariants. This establishes physical CP violation in a conditional matched example; it does not derive a unique phase or select the physical vacuum. Finally, the canonical cap connection is certified to have no zero modes. A fully specified modified global mass-squared operator has exactly three zero modes among 44,352 modes, with a gap of one half. Its required symmetry breaking and its chiral quark completion remain independent assumptions.

Exact CP on the elementary field content

$$\overline{X}R(g)X^{-1} = R(u(g)), \quad X\overline{X} = I$$

Use the published outer automorphism that sends the faithful triplet to its conjugate: a maps to a^2 , h to q , f to f^2 , h to q , and q to a . We evaluate these words on the exact generators, propagate the map through all 1080 elements, and check that every repeated group word has the same image. We solve $X \text{conjugate}(R(g)) = R(u(g)) X$ as exact linear equations over $Q(\sqrt{5}, \sqrt{-3})$. The published representative can square to an inner action. An explicitly recorded inner adjustment yields a symmetric unitary X with $X \text{conjugate}(X)=I$.

The receipt gives the matrix entries, its normalization and the adjustment word. Before normalization, $X_0^\dagger X_0 = (6-2\sqrt{5}) I$. The sextic identity is checked coefficient by coefficient: $F_6(X_0 \phi) = (6-2\sqrt{5})^3 \text{conjugate}(F_6)(\phi)$. Equivalently $F_6(X \text{conjugate}(\phi)) = \text{conjugate}(F_6(\phi))$. Thus real coefficients of F_6 plus its conjugate define an invariant interaction on this specified field content. Ordinary componentwise conjugation in the original basis would not establish this result.

There is an important scope restriction. We enumerate all 17 conjugacy classes and the four outer-class representatives supplied by the known Klein outer automorphism group. Every representative fails to send at least one class to its inverse class. For the representative interchanging this triplet with its conjugate, two classes fail. Inner adjustments do not change these class permutations. There is therefore no class-inverting automorphism that could define the same physical CP operation for arbitrary irreducible field content. The anti-linear operation constructed here is restricted to the elementary triplet, singlet and bifundamental fields that pass the explicit consistency and Lagrangian checks. Adding a different elementary representation requires a fresh audit.

Primary sources: Hagedorn, Meroni and Vitale, <https://arxiv.org/abs/1307.5308>, equation (34), for the outer maps and their classification; Chen et al., <https://arxiv.org/abs/1402.0507>, for the class-inversion requirement in generic finite-group CP; <https://doi.org/10.1140/epjc/s10052-025-15050-8>, for the representation-content dependence of

conjugation in Sigma(1080). Our exact class enumeration and intertwiner calculation are independent computations using the recovered generators.

Cross-sector sequestering and its exact operator boundary

$$G_u \times G_d, \quad |\phi_{ui}^\dagger \wedge \phi_{dj}|^2 \quad (\text{degree } 6)$$

Take ϕ_{ui} in (3,1), ϕ_{dj} in (1,3), and Λ in (3,3 $\bar$). Retain one independent C6 shaping factor for each right-handed singlet and its flavon. Before a link is inserted, $|\phi_{ui}^\dagger \phi_{dj}|$ squared is not an invariant of the product family group. There is no invariant identification of its independent family indices. Its proposed replacement $|\phi_{ui}^\dagger \Lambda \phi_{dj}|$ squared is invariant and has field degree six. Its nine coefficients remain independent. Arbitrary additional phase charges could not have forbidden the original absolute overlap square in the diagonal-group model: its charges cancel identically.

The bifundamental Molien calculation averages over the 1080 squared product group using exact power-trace class signatures and Newton identities. Through total degree six it has nonzero spaces (1,1) of dimension one, (3,0) and (0,3) of dimension one, (2,2) of dimension two, (4,1) and (1,4) of dimension one, (3,3) of dimension three, and (6,0) and (0,6) of dimension two. This is the complete self-operator count through that degree, not only a selected contraction.

The degree-three invariant is $\det \Lambda$. The two holomorphic degree-six invariants are its square and $I_6(\Lambda)$, obtained by contracting the unique symmetric sextic tensor with six link matrices and its conjugate tensor. Consequently there is only one genuinely new holomorphic orientation tensor at this degree. Balanced link-only sextics are the three symmetric polynomials of the singular values of Λ . At degree five the invariant is $\det \Lambda$ times $\text{Tr}(\Lambda \dagger \Lambda)$. Real parts are selected by the field-content CP involution; none of these counts supplies a relation between independent coupling constants.

The complete renormalizable scalar count for the listed six flavons, one link and one Higgs is 58 real parameters with this CP restriction. There are eight quadratic norms, one real cubic determinant coefficient, and 49 quartics. The quartics consist of 21 flavon norm products, six within-sector absolute overlaps, two link self-contractions, six link-norm/flavon-norm products, six $\phi \dagger \Lambda \Lambda \dagger \phi$ or conjugate-side contractions, and eight Higgs quartics and norm portals. The nine direct up/down overlap quartics of the earlier diagonal-group model are absent. Without CP the determinant coefficient has two real components, giving 59 parameters. The counts refer to this explicitly assumed product family and shaping content, not to every possible UV theory.

On a unitary link, $\Lambda \dagger \Lambda$ is proportional to I. The six link/flavon orientation quartics then become norm terms and cannot rotate a source frame. Thus this construction removes the previous quartic angular perturbation in that regime. At degree six the dressed overlaps return, and a general Wilsonian effective theory permits them even if a chosen tree potential sets their coefficients to zero. They must be controlled by an additional mechanism before an exact CKM prediction could be protected.

A representation-derived positive link interaction

$$I_6(U) = \langle T, T \circ U \rangle, \quad \text{Re } I_6(U) \leq \|T\|^2 = 16$$

Let T be the symmetric coefficient tensor defined by $F_6(\phi) = T$ contracted with six copies of ϕ . Tensor norms require multinomial weights; the exact polynomial coefficients give $\text{norm}(T)$

squared=16. Define $I6(\Lambda) = \text{inner product of } T \text{ with } T \text{ composed with } \Lambda$. For unitary U , $\text{norm}(T \text{ composed with } U) = \text{norm}(T)$, so $\text{Re } I6(U)$ is at most 16 by Cauchy-Schwarz. Equality is equivalent to $T \text{ composed with } U = T$. All 1080 exact group matrices satisfy that tensor identity, independently of the chosen quark model or CKM target.

A globally nonnegative polynomial alignment energy is $V_L = a \text{Tr}[(\Lambda^\dagger \Lambda - \ell^2 I)^2] + b |\det \Lambda - \ell^3|^2 + c \text{norm}[T \text{ composed with } \Lambda - \ell^2 T]^2$, for positive a, b, c and ℓ . Every $\Lambda = \ell R(g)$ is an exact zero-energy state. On the unitary, determinant-fixed manifold its last term is proportional to $16 - \text{Re } I6(U)$. The last norm square contains degree-six and degree-twelve terms; the interaction is not a wholly renormalizable potential. Its coefficients and its higher-degree completion are explicit choices. We establish these zero-energy states and their positivity, without claiming an independently derived UV origin or a complete classification of all zero-energy matrices.

The link transforms under the elementary CP operation as Λ maps to $X \text{conjugate}(\Lambda) X^\dagger$. $I6$ and $\det \Lambda$ then map to their conjugates. The positive interaction is therefore invariant. It correlates the sectors through a fixed representation tensor rather than through an arbitrary fitted matrix. Its degenerate group-related states do not select a physical CKM phase by themselves.

A renormalizable messenger origin for the sextic

The sextic also has an explicit interaction-level origin. Symmetric squared of the faithful triplet is an irreducible complex six, and symmetric cubed is an irreducible real ten. The latter has a unique invariant symmetric bilinear Q_{10} . Polarizing the sextic across two groups of three indices gives this bilinear, with $F6(\phi) = \text{Sym}^3(\phi) \text{transpose } Q_{10} \text{Sym}^3(\phi)$. In an orthonormal symmetric-cube basis its exact identity is $Q_{10}^\dagger Q_{10} = (8/5) I$. We certify all 100 entries of that matrix identity in the coefficient field. This establishes nondegeneracy and fixes the internal component coefficients independently of a desired root.

For the product-family link introduce supersymmetric chiral messengers A in $(6, \bar{6})$, $A_{\tilde{}}$ in $(\bar{6}, 6)$, and B in $(10, 10)$, with dimensions 36, 36 and 100. These are Standard Model singlets and neutral under the six shaping factors. The superpotential chain contains $M6 A \cdot A_{\tilde{}}$, $M_{10} B \cdot B$ divided by two, $y_2 A \cdot (\Lambda \Lambda)$ projected to the 36, and $y_3 B \cdot (A_{\tilde{}} \Lambda)$ projected to the 100. Every displayed family contraction is unique. The CP action on these messenger fields is inherited from the symmetric-square and symmetric-cube actions of the already checked X ; no arbitrary new CP matrix is introduced.

Eliminating the massive fields at leading tree order generates a holomorphic operator proportional to $-(y_2 y_3)^2 I6(\Lambda) / (2 M_6^2 M_{10})$, in compatible invariant normalizations. Thus the sextic is generated by renormalizable cubic vertices whose representation coefficients fix its component pattern. This is stronger than simply inserting a high-degree polynomial into the scalar potential. It is a supersymmetric messenger-sector construction, not a completed softly broken scalar model: the F-term scalar potential from the effective superpotential, soft terms, canonical kinetic corrections and physical vacuum selection must be matched separately.

The full audit counts all 34 possible monomial multisets of degrees one through three for these four chiral field species. There are no invariant linear terms, two independent mass contractions, and 15 cubic contractions: the determinant of Λ ; the two unique chain vertices; and four self-cubic contractions each for A , $A_{\tilde{}}$ and B . The unused self-cubics are not silently excluded. $A_{\tilde{}}$ self-cubics can contribute at degree six to the same two-dimensional link-invariant space. The A and B self-cubics first affect higher orders in the

small-link expansion for this massive branch. Hence the generated orientation tensor is fixed, while its coefficient and the determinant-squared coefficient remain independently adjustable. Other branches and the regime of large link expectation values require the full messenger equations.

The extra messenger fields do not couple holomorphically to the charged source flavons at renormalizable degree under the listed shaping charges. This controls a concrete set of interactions, while higher-degree Kahler and effective scalar terms can still generate the dressed cross-sector overlaps. No radiative absence theorem for those overlaps is asserted.

Complete mediator matching with the reverse interaction included

Assign Q_L to (3,1). The up-sector mediator remains a triplet of G_u . For the down sector introduce D_u in (3,1) and D_d in (1,3), both vectorlike Standard Model weak singlets of down electric charge. The right-handed down singlets couple to D_d through the three down flavons. The allowed link Yukawa couplings run in both directions: $g \bar{D}_u L \Lambda D_d R$ and $g_{\text{reverse}} \bar{D}_d L \Lambda^\dagger D_u R$. Both are allowed by the same mass and shaping charges. Omitting the reverse contraction would incorrectly restrict the operator space.

At $H=0$ the down heavy block is $[B, M]$, with $B=[0; C_d]$ and $M=[M_u I, g \Lambda; g_{\text{reverse}} \Lambda^\dagger, M_d I]$. The light normalized right frame is $[I; -A] K$ to the power $-1/2$, where $A=M^{-1}B$ and $K=I+A^\dagger A$. The Higgs couples Q_L to $D_u R$, giving $Y_d=-h A_{\text{top}} K$ to the power $-1/2$. For an invertible M there are exactly three right light modes and six massive down mediator modes.

For $\Lambda=\ell U$ and real parameters, write $D=M_u M_d - g g_{\text{reverse}} \ell^2$. Then $A_{\text{top}}=-g \Lambda C_d/D$ and $A_{\text{bottom}}=M_u C_d/D$. Consequently $K=I+[(g^2 \ell^2 + M_u^2)/D^2] C_d^\dagger C_d$, and Y_d is a scalar times $U C_d$ times that canonical factor. The left frame is exactly U times the left singular frame of C_d . This includes the reverse link coupling and all orders in flavon mixing. The heavy masses must remain nonzero, requiring D nonzero. Finite-Higgs and quantum threshold corrections are separate calculations.

Physical CP established through quark invariants

$$J \neq 0, \quad \text{Im det}[H_u, H_d] \neq 0, \quad J_{\text{CP}} = -J$$

The numerical example uses a unitary W spanning the real fixed space of the anti-linear map, so $X \text{conjugate}(W)=W$. Its source matrices are $C_u=W \text{diag}(0.07, 0.31, 0.9)$ and $C_d=W \text{diag}(0.08, 0.27, 0.85)$. All listed mediator couplings are real. We evaluate every group link $U=R(g)$, select the largest absolute J without a target phase or mixing anchor, and calculate the conjugate link $X \text{conjugate}(U) X^\dagger$.

After canonical matching the pair has $|J|$ approximately 0.0703201663, identical mixing magnitudes and opposite J signs. The chart phases are approximately plus and minus 94.6111562 degrees, and C_{eff} is approximately 1.50313413. An independent weak-basis check gives nonzero $\text{Im det}[H_u, H_d]$, approximately $7.09e-9$ in absolute value. Nonzero values with nondegenerate quark spectra demonstrate physical CP violation; complex representation entries or flavon phases alone would not do so.

This example is conditional on the chosen CP-real source frames and on a selected group link vacuum. The CP-real basis itself has a real orthogonal freedom, and the scalar model has not

selected those frames. The large mixing is not phenomenologically viable. Real, independently allowed column-coupling changes in a nonorthogonal source example alter both C_{eff} and the physical phase. The receipt retains all cases. Thus physical CP is possible within a consistent restricted elementary symmetry, but the observed phase and the golden relation remain unpredicted.

The canonical global spectrum is gapped

The recovered six-dimensional real cap fiber is wedge squared of the standard four-dimensional A5 representation, with its stored positive metric. Construct the covariant graph Laplacian using the actual 252 directed incidence transports on a 42-cap component. In integral coordinates its characteristic polynomial is $(t-2)15 (t-3)55 (t-5)21 (t-6)70 (t-7)21 (t-9)55 (t-10)15$. We compute and factor this polynomial exactly over the integers. The numerical metric-whitened Hermitian matrix independently reproduces the spectrum.

The base connection has dimension 252 and zero-dimensional kernel. Its orientation double cover has dimension 504 and the same spectral levels with doubled multiplicities: the other block is obtained by t mapping to $12-t$. Both have exact mass-squared gap two. On all 176 isomorphic cap components the base space contains 44,352 modes and has no zero mode. A parallel rank-three projector therefore does not yield three parallel vectors. The A5 cover holonomy has no invariant vector in either irreducible golden triplet, consistent with the spectral result.

A global three-mode operator with explicit breaking

$$\dim \ker M^2 = 3, \quad \lambda_{\text{next}}(M^2) = \frac{1}{2}, \quad N = 44\,352$$

To exhibit exactly what an additional assumption can accomplish, choose one 42-cap component and a spanning tree, transport a root frame along that tree, and replace every edge matrix by the corresponding pure-gauge transport $R_w R_v$ inverse. This keeps the tree transports and changes 160 of the 252 directed edge matrices of the component. It trivializes the holonomy and is an explicit change to the canonical connection. It is not invariant under the full recovered M22 incidence action.

Transport the root rank-three Hodge projector by the same frames, and add a one-half penalty on its orthogonal complement. The scalar component Laplacian has exact spectrum 0 with multiplicity one, 4 with multiplicity 21, 7 with multiplicity six, and 9 with multiplicity 14. The selected six-dimensional bundle therefore has three zero modes, and levels one half, 4, 4.5, 7, 7.5, 9 and 9.5 with multiplicities 3, 63, 63, 18, 18, 42 and 42 respectively. Gap every other cap fiber by an onsite mass squared of two, adding 44,100 massive modes.

This explicitly defined global mass-squared operator has exactly three zero modes among 44,352 modes and gap one half. Their normalized wavefunctions are transported root vectors divided by $\sqrt{42}$; the numerical mass and normalization residuals are below $2e-15$. The spectral statement is exact from the scalar characteristic polynomial and pure-gauge similarity. Selecting the component breaks M22, and flattening its connection breaks further symmetry. We have not derived either selection dynamically. The operator is a positive Hermitian flavor mass-squared construction, not a gauge-invariant chiral quark UV completion. Its real A5 transport and the local complex Valentiner link model have not been unified into one global particle theory.

What remains after this push

The bare quartic deformation is now excluded by specified product symmetry, but the degree-six dressed operators survive. The CP action is explicit on the listed fields, and physical CP is demonstrated in canonically matched matrices, but its vacuum and numerical phase are not selected. The global spectrum is calculated, including an exact obstruction for the canonical connection and a conditional three-mode alternative, but the required breaking interactions and chirality have not been derived. These boundaries are sharper than another constructed golden root. A complete predictive theory must connect the product-family link, its mixed operator coefficients and CP-compatible scalar vacua to a dynamically selected global chiral spectrum.

Reproduction and receipts

Run `python python/develop_valentiner_cp.py`, `python python/develop_valentiner_link.py` and `python python/develop_m22_global_modes.py`, and `python python/develop_valentiner_messenger.py`. The computations use the existing exact triplet generators, sextic receipt and frozen cap transports. The new receipts are `valentiner_cp.json`, `valentiner_link.json`, `m22_global_modes.json` and `valentiner_messenger.json` under `receipts/m22_interactions`. The CP and polynomial identities, class map propagation, bifundamental invariant counts and characteristic-polynomial factors use exact arithmetic. The quark examples, matrix whitening and normalization residuals are numerical cross-checks. No experimental data or new Lean theorem is claimed.